NON-LOCAL SPARSE ATTENTION BASED MULTI-IMAGE SUPER-RESOLUTION

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***Abstract*—Modern satellites often revisit the same region with high frequency, providing an opportunity for the use of Multi-Image Super Resolution (MISR) in remote sensing. However, temporal variations and weak inter-image correlations make effective utilization of multiple low-resolution observations challenging. While recent transformer-based methods improve multi-image fusion, attention mechanisms are typically applied only during fusion or reconstruction, leading to information loss during feature encoding. In this paper, we propose an enhanced transformer-based MISR architecture that integrates non-local sparse attention (NLSA) into the encoding stage to capture long-range spatial dependencies across images. We demonstrate that explicitly capturing spatial dependencies through NLSA during the encoding stage enables random shuffling of the image sequence during training to be effective, leading to improved utilization of multi-image information and overall model performance. We also introduce a novel combination of loss functions that increases texture preservation and structural integrity. Experiments on the PROBA-V Kelvin dataset demonstrate that the proposed method outperforms state-of-the-art approaches.**

***Index Terms*—Super-Resolution, Multi-image Superresolution, Computer Vision, Remote Sensing**

I. Introduction

Satellite imagery has become an indispensable tool for remote sensing, enabling frequent and global observations. However, due to cost and data transmission rate constraints, satellites are often unable to capture large volumes of high-resolution (HR) imagery [1], limiting its application to practical needs, including disaster monitoring [2], land cover mapping [3], and vegetation growth analysis [4]. To overcome this limitation, various studies [1, 5, 6] have proposed image super-resolution (SR) techniques, demonstrating their viability in the remote sensing (RS) domain.

Image SR is an advanced computer vision technique that enhances the resolution and details of low-resolution (LR) images. Researchers have categorized image SR techniques into single-image SR (SISR) [7] and multi-image SR (MISR) [8]. MISR methods merge multiple images collected of the same scene to reconstruct an HR image. In remote sensing, MISR faces challenges because images are often captured over long time spans, leading to uncontrolled environmental changes. As a result, efficient fusion of multiple low-resolution images remains difficult under these weak temporal correlations.

Traditional approaches for MISR rely on prior knowledge (e.g., phase shift or geometric distortion) to reconstruct an HR image from LR images. Tsai et al. [8] reconstructed an HR image by fusing LR images in the Fourier domain. Other traditional approaches apply iterative optimization frameworks incorporating explicit image priors [9–13]. However, this assumption on prior knowledge, limits the applicability of these methods in real-world scenarios. In contrast, deep learning approaches, particularly convolutional neural networks (CNNs), have shown promising results. The first of these approaches, HighRes-net [6], is an end-to-end encoder-decoder architecture that encodes multiple LR images into feature maps, which are recursively fused. The decoder then decodes the fused features to obtain an HR image. Other methods improve the encoding process by convGRU [14], 3-D convolutions [15] and graph convolution [16], while still relying on recursive fusion. To address the challenges of uncertainty and temporal variations in image sequences, some studies [17, 18] have incorporated random shuffling during training; however, these models lack sequence-level contextual information. TR-MISR [19] uses transformer layers to dynamically generate attention weights for images in the sequence with contextual information. However, this method ignores image spatial dependencies, limiting its ability to capture larger receptive fields. To address this limitation, Ding et al. proposed AttMISR [20], which uses hybrid non-local fusion and captures spatial dependencies during the reconstruction process through coordinate-aware attention. However, this approach has two limitations: (1) No attention mechanism is employed during the encoding process, which limits the model's capacity to leverage multi-image information, leading to information loss; and (2) the hybrid non-local fusion and coordinate-window attention mechanisms are computationally expensive. Furthermore, these methods optimize the mean square error (MSE), which is known to favor pixel-wise averaging and often leads to overly smoothed reconstructions and reduced perception quality [21, 22].

We have modified the TR-MISR model to overcomes these limitations in MISR. Our main contributions are as follows: (1) An enhanced TR-MISR architecture that uses non-local sparse attention (NLSA) [23] during the encoding process to efficiently capture a larger receptive field. (2) A new loss function for MISR that combines MSE loss with wavelet loss [5] to improve reconstruction quality. (3) We demonstrate that capturing spatial dependencies (with NLSA) during the encoding stage makes random shuffling of the image sequence during training effective in improving the overall model performance. Our modified approach outperforms state-of-the-art methods on the PROBA-V Kelvin dataset.

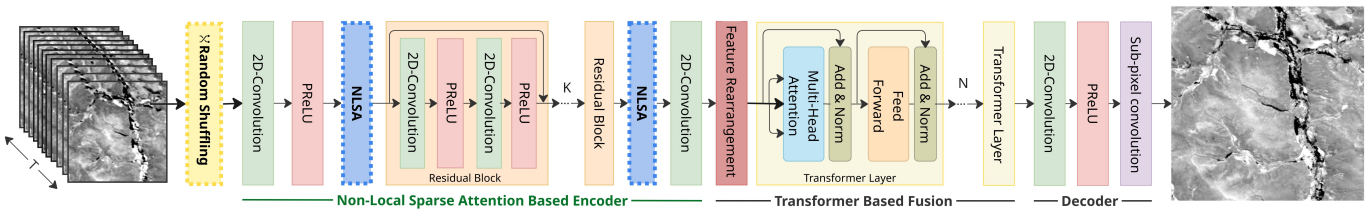


Fig. 1. Proposed MISR architecture that modifies TR-MISR [19] with NLSA [23] to capture long-range dependencies more effectively and preserve high-frequency information representation, respectively. Modification to baseline TR-MISR [19] (NLSA and random shuffling) are highlighted with a dotted border.

II. Methodology

A. Network Architecture

Our approach enhances the TR-MISR architecture [19] by efficiently improving its capacity to capture long-range dependencies (see Fig. 1 for an architecture overview). TR-MISR consists of three parts: a residual-based encoder block, a transformer-based fusion block and a decoder block. The encoder block refers to Highres-net [6], which first applies a 2D convolutional layer and a PReLU activation function [24] to extract shallow features, then K residual blocks are applied to derive deeper features. Each residual block consists of two 2D convolutional layers, each followed by a PReLU activation. We improved the residual blocks of TR-MISR by interposing them between the NLSA [23] block to overcome the limitations of the convolutional layer, with respect to the limited receptive field, thereby efficiently capturing long-range dependencies. NLSA employs spherical locality-sensitive hashing to group features into buckets and perform sparse attention. This mechanism enables the model to focus on informative regions without paying attention to unrelated regions. By approximating global non-local attention through sparse feature interactions, NLSA effectively captures long-range dependencies. Moreover, hashing-based sparsity significantly reduces computational complexity compared to dense attention mechanisms. Afterwards, the multiple feature maps are rearranged into vectors, enabling the fusion module to dynamically assign attention to corresponding spatial regions across different images. In addition, TR-MISR introduces a learnable embedding vector that aggregates these feature vectors, allowing the model to fuse multi-image information effectively and restore fine details. A convolution operation followed by PReLU is applied to reduce the channel dimension of the feature maps and obtain a response map. Finally subpixel convolution [25] is applied to produce the output.

B. Loss Function

The modified Transformer-based MISR architecture is trained end-to-end using supervised learning. Given multiple low-resolution images $\{\mathrm{LR}_i\}_{i=1}^{T}$ as input, the network generates a single super-resolved image SR, while the corresponding high-resolution image HR serves as the ground truth. Due to different collection environments of the low-resolution images and the ground truth, vertical and horizontal offsets as well as an overall brightness deviation may exist between SR and HR. We assume the alignment error is the same across the whole image and at most c pixel in both directions. We then compensate for the offsets using a center-crop of the predicted image by c pixels to obtain $\mathrm{SR}_{c,c}$. The ground truth image is cropped to the same spatial size using a sliding window with starting coordinate (u, v), producing $\mathrm{HR}_{u,v}$, where $u, v \in \{0, \ldots, 2c\}$ in order to find the best fitting alignment between $\mathrm{SR}_{c,c}$ and $\mathrm{HR}_{u,v}$. A binary mask $\mathbf{M}_{u,v}$ indicates the valid pixels of $\mathrm{HR}_{u,v}$.

Following the same methodology as previous works [19, 20], the average brightness bias between $\mathrm{SR}_{c,c}$ and $\mathrm{HR}_{u,v}$ is computed over the valid pixels as

$$b_{u,v} = \frac{\sum_{x,y}(\mathrm{HR}_{u,v}(x,y) - \mathrm{SR}_{c,c}(x,y)) \odot \mathbf{M}_{u,v}(x,y)}{\|\mathbf{M}_{u,v}\|_1},$$

Where $\odot$ is the Hadamard product. We define masked and brightness adjusted images $\tilde{\mathrm{SR}}_{u,v}$ and $\tilde{\mathrm{HR}}_{u,v}$ as

$$\tilde{\mathrm{SR}}_{u,v} := (\mathrm{SR}_{c,c} + b_{u,v}) \odot \mathbf{M}_{u,v}, \quad \tilde{\mathrm{HR}}_{u,v} := \mathrm{HR}_{u,v} \odot \mathbf{M}_{u,v}.$$

The mask-normalized MSE loss (selecting the best spatial shift) is

$$\mathcal{L}_{\mathrm{MSE}} = \min_{u,v} \frac{\|\tilde{\mathrm{SR}}_{u,v} - \tilde{\mathrm{HR}}_{u,v}\|_2^2}{\|\mathbf{M}_{u,v}\|_1}.$$

This classic per-pixel Euclidean distance loss assumes pixel-wise independence, inadequately capturing high-frequency details and structured image content, leading to blurry outputs that misalign with human perception [5, 22, 21]. We address this limitation by combining MSE RGB loss with a wavelet loss, which decomposes images into multi-scale subbands, enabling detailed reconstruction of a predicted image. Previous work has shown the effectiveness of this wavelet loss for SISR [5]. However, its effectiveness in MISR is still unexplored. Specifically, we use an SWT (Stationary Wavelet Transform) loss [26] (selecting the best spatial shift) in addition to pixel-wise MSE loss for MISR. The SWT loss is computed as

$$\mathcal{L}_{\mathrm{SWT}} = \min_{u,v} \sum_j \lambda_j \left\| \mathrm{SWT}_j\left(\tilde{\mathrm{SR}}_{u,v}\right) - \mathrm{SWT}_j\left(\tilde{\mathrm{HR}}_{u,v}\right) \right\|_2^2,$$

where the sum is taken over wavelet sub-bands j and λ_j controls each sub-band's contribution.

The final training objective is given by

$$\mathcal{L}_{\mathrm{total}} = \mathcal{L}_{\mathrm{MSE}} + \mathcal{L}_{\mathrm{SWT}}.$$

TABLE I
QUANTITATIVE COMPARISON ON THE PROBA-V DATASET USING CPSNR AND CSSIM METRICS.

Method	NIR		RED		ALL	
	cPSNR	cSSIM	cPSNR	cSSIM	cPSNR	cSSIM
Bicubic	45.44	0.9770	47.33	0.9840	46.40	0.9806
BTV[10]	45.93	0.9794	48.12	0.9861	47.04	0.9828
IBP[27]	45.96	0.9796	48.21	0.9865	47.10	0.9831
CARFT-SR[28]	45.09	0.9595	47.36	0.9707	46.24	0.9652
ATD[29]	45.11	0.9593	47.40	0.9705	46.27	0.9649
DAT[30]	45.09	0.9595	47.36	0.9707	46.24	0.9652
RCAN[31]	45.44	0.9770	47.33	0.9840	46.40	0.9806
VSR-DUF[32]	47.20	0.9850	49.59	0.9902	48.42	0.9876
HighRes-Net[6]	47.55	0.9855	49.75	0.9904	48.67	0.9880
3DWDSRNet[17]	47.58	0.9856	49.90	0.9908	48.76	0.9882
DeepSUM[15]	47.84	0.9858	50.00	0.9908	48.94	0.9883
MISR-GRU[33]	47.88	0.9861	50.11	0.9910	49.01	0.9886
PIUnet[34]	48.72	0.9883	50.62	0.9921	49.68	0.9902
RAMS[18]	48.17	0.9869	50.13	0.9911	49.17	0.9890
RAMS_{+20}[18]	48.27	0.9870	50.27	0.9912	49.29	0.9891
TR-MISR[19]	48.54	0.9882	50.67	0.9921	49.62	0.9902
AttMISR[20]	48.94	0.9883	50.64	0.9923	49.79	0.9903
Ours	**49.15**	**0.9884**	**51.15**	**0.9924**	**50.20**	**0.9905**

III. EXPERIMENTS & RESULTS

A. Dataset

All our experiments use the popular PROBA-V dataset [35] which is a commonly used benchmark for satellite-imagery MISR [15, 17–20, 33, 34]. It provides data for 74 manually selected regions consisting of a red and NIR band at $100\mathrm{m}$ and $300\mathrm{m}$ resolution respectably. In total 1450 scenes are provided with an average of 19 different LR images per scene (minimum is 9), collected within a time window of 30 days. The LR images are provided in 128×128 pixel, the HR images in 384×384, which results in a scaling factor of 3. The images have a bit-depth of 14 but are provided as 16-bit images. A quality map is provided for each image, providing information on pixel-wise concealment values. For the image sequences we follow the same procedure as [19] selecting $T = 24$ images per sequence (shorter sequences are padded by blank images).

B. Evaluation Metric

To accurately evaluate the experimental results, we employed cPSNR[13] and cSSIM[12], which are specifically designed for the MISR task. These evaluation metrics overcome the problems of image offsets and brightness deviation in the MISR task and are widely used in the PROBA-V challenge. The cPSNR is computed as:

$$\mathrm{cPSNR} \;=\; \max_{u,v} \; 10 \log_{10}\!\left(\frac{(2^{16}-1)^2}{\mathrm{MSE}_{u,v}}\right),$$

where

$$\mathrm{MSE}_{u,v} \;=\; \frac{\left\| \tilde{\mathrm{HR}}_{u,v} - \tilde{\mathrm{SR}}_{u,v} \right\|_2^2}{\|M_{u,v}\|_1},$$

represents the mean squared error computed on clear pixels indicated by the binary mask $M_{u,v}$ $2^{16}-1$ denotes the maximum pixel intensity for an image encoded on 16 bits. Since both SR and HR are 384×384 pixels with the maximum registration offset $c = 3$, the starting coordinates are $u, v \in \{0, \ldots, 6\}$.

Similarly,

$$\mathrm{cSSIM} \;=\; \max_{u,v} \; \mathrm{SSIM}\big(\tilde{\mathrm{HR}}_{u,v},\, \tilde{\mathrm{SR}}_{u,v}\big),$$

computes the maximum SSIM between $\tilde{\mathrm{SR}}_{u,v}$ and $\tilde{\mathrm{HR}}_{u,v}$.

C. Implementation Details

We have used two NLSA [23] blocks with their default configuration, added before and after the residual blocks of TR-MISR [19]. For end-to-end training, we followed the default hyperparameter settings of TR-MISR [19], setting the number of residual blocks to 2, the number of transformer heads to 8, and the number of transformer layers to 6. We performed random shuffling during training and set the batch size to 8. In contrast to TR-MISR, we trained our model on both bands and performed inference on NIR, RED, and the two bands combined. The original TR-MISR architecture required a fine-tuning step in order to achieve the best result, i. e. another 20 epochs with a reduced learning rate of $5 \cdot e^{-5}$ for the encoder and $2.5 \cdot e^{-5}$ for the fusion module. We found this to be unnecessary for our modified version. Our model's optimization is faster than previous works (which were trained for 400 epochs) [18–20] achieving the best results with only 250 epochs. All experiments are performed on a single Nvidia A6000 GPU. All code and pre-trained models are provided in our GitHub repository[1].

D. Performance Evaluation

In this section, we discuss the quantitative and qualitative performance of our proposed approach. We have conducted a quantitative evaluation on two widely used metrics for the MISR task (i.e., cPSNR and cSSIM). As reported in Table I, our proposed approach performs best on the RED and NIR bands, and across all bands. Additionally, we present the qualitative results in Fig. 2. The top half of each row shows the bicubic interpolation of the LR image, the predictions of the two models with released code and weights (i.e., RAM+20 and TRMISR), and our model. For each image, we have highlighted some regions to illustrate the structural reconstruction capabilities of our model. The bottom half of both rows represents the absolute difference in pixels between the predicted and true values (i.e., error maps) to show pixel-wise reconstruction capabilities. In the error map, the closer the pixels are to white, the smaller the difference from the actual value and the better the reconstruction results; conversely, the further away from white the pixels are, the larger the difference and the worse the reconstruction results. We also use the same center crop strategy to compensate for pixel offsets in the computation of the error maps. Our model achieves the best reconstruction performance, which showcases the effectiveness of our proposed approach. Our experiments have also shown that training for only 100 epochs with our model is sufficient to beat the baseline model (the model then improves and converges at 250 epochs).

[1] github.com/TBD

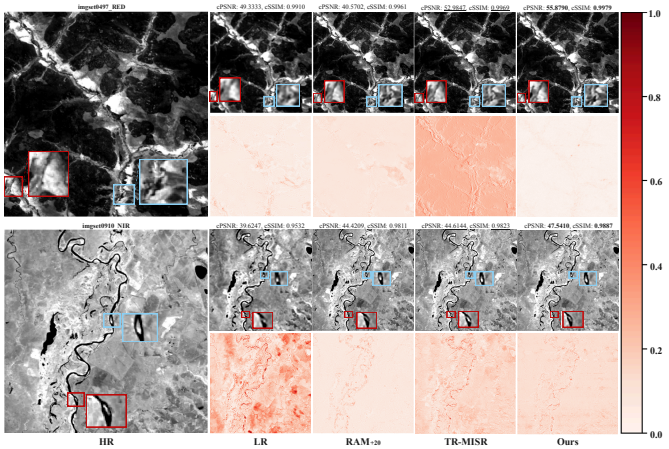


Fig. 2. Comparison of High-Resolution Ground Truth (HR) images from Low-Resolution (LR Input) with the outputs (RAM+20 [18], TR-MISR [19]), and the proposed model (Ours). We enlarged the regions outlined in red to show the performance of each model in reconstructing fine details. Bottom row of each image represents pixel-wise absolute error maps. The colors go from white (low error) to red (high error). The error is scaled normalized between 0 and 1 using the maximum absolute error over all methods per image.

TABLE II
COMPUTATIONAL COMPLEXITY AND PERFORMANCE COMPARISON OF BASELINE (TR-MISR) WITH DIFFERENT NUMBERS OF NLSA BLOCKS.

Model	FLOPs (G)	Params (M)	cPSNR	cSSIM
Baseline	**38.71**	**0.386**	49.62	0.9902
+1 NLSA	46.83	0.466	50.13	0.9903
+2 NLSA	48.14	0.479	**50.20**	**0.9905**
+3 NLSA	49.45	0.492	49.51	0.9901

E. Ablation Studies

We perform various ablation studies to show the effectiveness of (1) the inclusion of NLSA blocks, (2) wavelet loss, and (3) random shuffling on NLSA-based feature maps. All ablations are conducted on all bands, and results are reported using cSSIM and cPSNR metrics.

Table II depicts the increase in complexity and memory usage of our model through the inclusion of the NLSA blocks. We achieve the best results with the inclusion of two NLSA blocks which help to improve the cPSNR score especially. This comes at a cost of $\approx 24\%$ more FLOPs and parameters but the model still remains relatively compact. The inclusion of a third NLSA block does not help and in fact performs even worse than the baseline suggesting that the benefit of including more attention blocks is exhausted and the additional parameters are counterproductive. As reported in Table III, we analyze the effectiveness of different loss functions in the proposed approach. When the model is optimized using only the MSE loss, the performance remains limited due to its tendency to favor pixel-wise averaging. By incorporating the SWT loss together with MSE, a clear improvement in cPSNR is observed, as the SWT loss enhances reconstruction quality in the frequency domain by preserving high-frequency details. We further evaluate the impact of the NLSA module trained only with the MSE loss and observe a minor performance improvement, attributed to its enlarged receptive field. The best performance is achieved with the NLSA module optimized with both $\mathcal{L}_{\mathrm{MSE}}$ and $\mathcal{L}_{\mathrm{SWT}}$, demonstrating that the SWT loss becomes more effective when guided by attention-enhanced feature representations, allowing the model to exploit richer contextual information for improved reconstruction quality.

TABLE III
ABLATION STUDY WITH DIFFERENT LOSS FUNCTIONS

$\mathcal{L}_{\mathrm{MSE}}$	$\mathcal{L}_{\mathrm{SWT}}$	+NLSA	cPSNR	cSSIM
✓			49.62	0.9902
✓	✓		49.70	0.9899
✓		✓	50.16	0.9903
✓	✓	✓	**50.20**	**0.9905**

TABLE IV
ABLATION STUDY ON THE EFFECT OF NON-LOCAL SPARSE ATTENTION AND RANDOM SHUFFLING ON RECONSTRUCTION PERFORMANCE.

Baseline	+NLSA	+Random Shuffling	cPSNR	cSSIM
✓			49.62	0.9902
✓		✓	49.71	0.9899
✓	✓		50.01	0.9901
✓	✓	✓	**50.20**	**0.9905**

The effectiveness of random shuffling is analyzed in Table IV. Applying random shuffling to the baseline model results in a minor improvement in cPSNR, indicating limited benefits when performed on locally encoded features. In contrast, the best performance is achieved when random shuffling is applied to NLSA-encoded feature maps. This behavior can be explained by the non-local sparse attention module ability to capture large receptive fields and encode global contextual dependencies. Consequently, the shuffled images provide more informative and diverse inputs to the transformer-based fusion block mitigating information loss and enabling more effective multi-image fusion, thereby improving overall reconstruction performance.

IV. CONCLUSION & FUTURE WORK

In this paper we have introduced an extension to the popular TR-MISR network which achieves state-of-the-art performance on the prominent PROBA-V challenge for remote sensing MISR. This modified version (1) includes NLSA blocks into the model and (2) uses an additional wavelet-based loss function and is (3) trained with randomly shuffled image sequences. We provide ablations for all the inclusions which show that each inclusion is justified and increases relevant metrics. The experiments show that the inclusion of attention during the encoding process provides a significant benefit and that the wavelet-loss provides a tangible enhancement especially for the cPSNR metric.

Future work will incorporating temporal guidance, domain adaptation (such as forestry regions) using corresponding metrics (such as NDVI) and a new attention mechanism which spans between spatial information and contextual image sequence information.

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